

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&C) (Credit System) Degree Examinations

May - June 2023

20EC602 / 17EC501 / 15EC502 – DIGITAL COMMUNICATION

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

	Marks	BT*	CO*	PO*
1. a) With a neat diagram, explain the blocks in a basic digital communication system.	6	L*2	1	1
b) A signal $g(t) = 10\cos(20\pi t)\cos(200\pi t)$ is sampled at the rate of 250 samples/second.				
i. Sketch the spectrum of the sampled signal.				
ii. Specify the cutoff ideal reconstruction filter so as to recover $g(t)$ from $g_\delta(t)$.				
iii. Specify the Nyquist rate for the signal $g(t)$.	10	L3	1	1
c) Determine the Nyquist rate and the Nyquist sampling interval for the following signals.				
i. $g_1(t) = \text{sinc}(100\pi t)$				
ii. $g_2(t) = \text{sinc}^2(100\pi t)$	4	L3	1	1
2. a) With suitable diagrams and mathematical analysis, explain flat top sampling procedure.	10	L2	1	1
b) Draw the neat block diagram of time-division multiplexing system.	3	L1	1	1
c) A signal $m_1(t)$ is band-limited to 3.6 kHz, and three other signals $m_2(t)$, $m_3(t)$, and $m_4(t)$ are band-limited to 1.2 kHz. These signals are to be transmitted by means of TDM.				
i. Set up a scheme for realizing this multiplexing requirement, with each signal sampled at its Nyquist rate.				
ii. What must be the speed of the commutator in samples/second?				
iii. Determine the minimum bandwidth of the channel.	7	L3	1	1

Unit – II

3. a) Three signals $s_1(t)$, $s_2(t)$, and $s_3(t)$ are as shown in Fig.Q.3.a. Apply Gram-Schmidt procedure to obtain an orthonormal basis for the signals. Express signals $s_1(t)$, $s_2(t)$, and $s_3(t)$ in terms of orthonormal basis functions. Also, give the signal constellation diagram.

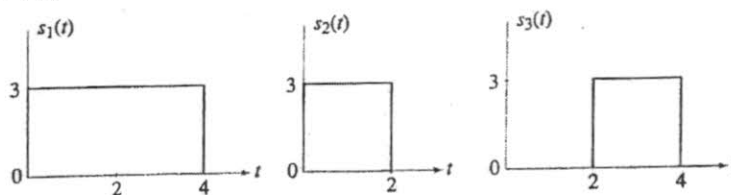


Fig.Q.3.a.

b) With suitable mathematical analysis, show that matched filter receiver is an optimum receiver.	6	L3	2	1
c) Draw diagrams for the detector and vector receiver parts of the correlation receiver.	10	L2	2	1
	4	L1	2	1

4. a) Derive an expression for the average probability of symbol error in a PCM system.
- b) For the sinusoidal modulating signal $x(t) = A_0 \cos(2\pi f_0 t)$, show that the output signal-to-quantization noise ratio in a delta modulated system under the assumption of no slope overload is given by

$$[SNR]_0 = \frac{3f_s^3}{8\pi^2 f_0^2 f_M}$$

where, $f_s = \frac{1}{T_s}$ = sampling frequency and f_M = cutoff frequency of the LPF in the receiver.

- c) For a binary PCM signal, determine L if the compression parameter $\mu = 100$ and the minimum $[SNR]_{0,dB} = 45 \text{ dB}$. Determine the $[SNR]_{0,dB}$ with this value of L.

10 L2 2 1

6 L2 2 1

4 L3 2 1

Unit – III

5. a) Derive an expression for the power spectral density of NRZ bipolar format.
- b) With a neat diagram of the baseband binary data transmission system, explain intersymbol interference.
- c) Show that the time response of a raised cosine spectrum with unity roll-off factor is given by;

$$p(t) = \frac{\text{sinc}(4B_0 t)}{1 - 16B_0^2 t^2}$$

7 L2 3 1

8 L2 3 1

5 L2 3 1

6. a) With relevant diagrams and mathematical analysis, explain duobinary encoding and decoding.
- b) An audio signal of bandwidth 4 kHz is sampled at a rate 25% above the Nyquist rate and quantized. The quantization error does not exceed 0.1% of the signal peak amplitude. The resulting samples are now encoded and transmitted by 4-ary pulses.
- Determine the minimum number of 4-ary pulses required to encode each sample.
 - Determine the minimum transmission bandwidth required to transmit this data with zero ISI.
- c) The binary data stream 011100101 is applied to the input of a correlative coder which has a correlation span of two binary digits.
- Construct the coder output and receiver output with a precoder. Assume $b_1=1$, $b_2=1$.
 - Suppose that owing to error during transmission, the level at the receiver input produced by the third digit is reduced to zero. Construct the new receiver output.

10 L2 3 1

4 L3 3 1

6 L3 3 1

Unit – IV

7. a) Derive an expression for the average probability of error for a coherent binary PSK system.
- b) An FSK system transmits binary data at a rate of 10^6 bits per second. Assuming channel noise is additive white Gaussian with zero mean and power spectral density 2×10^{-20} watts/Hz, determine the average probability of error. Assume coherent detection, and amplitude of received sinusoidal signal for both symbol 1 and 0 to be $1.2 \mu V$.

10 L2 4 1

5 L3 4 1

- c) A binary data is transmitted using ASK over an AWGN channel at a rate 2.4 MBPS. The carrier amplitude at the receiver is 1 mV. Noise power spectral density is 10^{-15} W/Hz. Find the average probability of error if the detection is (i) coherent, (ii) noncoherent. Take $\text{erfc}(5) \cong 3 \times 10^{-6}$.

5 L3 4 1

8. a) In a QPSK modulation scheme, the input binary sequence $\{b_i\} = \{11001011101\}$. The bit interval is T_b . Give the demultiplexed sequence $\{b_{ei}\}$ and $\{b_{oi}\}$. Assuming NRZ polar format, sketch the sequences $\{b_i\}$, $\{b_{ei}\}$ and $\{b_{oi}\}$.
- b) With neat diagrams, explain the transmitter and the receiver of differential phase shift keying system.
- c) Derive an expression for the average probability of error for M-ary QAM.

5 L3 4 1

8 L2 4 1

7 L2 4 1

Unit – V

9. a) Generate the maximum length sequence for the sequence generator given in Fig.Q.9.a, and verify the properties of the same. Assume the initial sequence of the shift register as 1010.

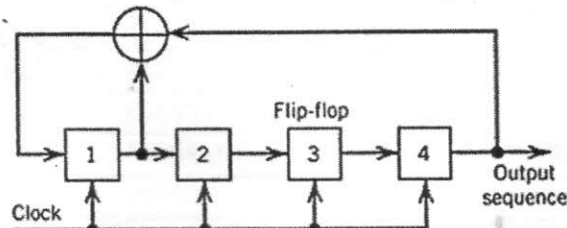


Fig.Q.9.a

- b) With a suitable model of DSSS system, derive an expression for the processing gain.

8 L3 5 1

12 L2 5 1

10. a) Bring out the differences between direct sequence and frequency hopped spread spectrum techniques.
- b) With neat diagrams, explain slow frequency hopped spread spectrum scheme.
- c) A slow FH/MFSK signal has the following parameters.

4 L2 5 1

8 L2 5 1

Number of bits per MFSK symbol $K = 2$ Number of MFSK tones $M = 2^K = 4$ Length of PN segment per hop $k = 3$ Total number of frequency hops $2^k = 8$

Illustrate the variation of the frequency of FH/MFSK signal for one complete period of the PN sequence. Assume that the carrier hops to a new frequency after transmitting two MFSK symbols or four information bits. Assume binary data sequence to be 10001101000111111001 and PN sequence to be 000100110101111. Also, sketch the variation of dehopped frequency with time.

8 L3 5 1

Table of Error Function

u	$\text{erf}(u)$	u	$\text{erf}(u)$
0.00	0.00000	1.10	0.88021
0.05	0.05637	1.15	0.89612
0.10	0.11246	1.20	0.91031
0.15	0.16800	1.25	0.92290
0.20	0.22270	1.30	0.93401
0.25	0.27633	1.35	0.94376
0.30	0.32863	1.40	0.95229
0.35	0.37938	1.45	0.95970
0.40	0.42839	1.50	0.96611
0.45	0.47548	1.55	0.97162
0.50	0.52050	1.60	0.97635
0.55	0.56332	1.65	0.98038
0.60	0.60386	1.70	0.98379
0.65	0.64203	1.75	0.98667
0.70	0.67780	1.80	0.98909
0.75	0.71116	1.85	0.99111
0.80	0.74210	1.90	0.99279
0.85	0.77067	1.95	0.99418
0.90	0.79691	2.00	0.99532
0.95	0.82089	2.50	0.99959
1.00	0.84270	3.00	0.99998
1.05	0.86244	3.30	0.999998

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
